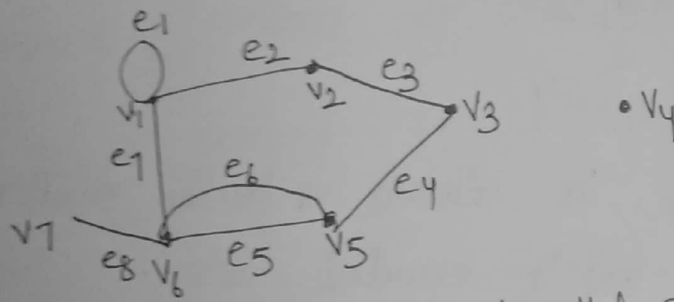


Graph theory

A graph $G(V, E)$ consists of a set of vertices denoted by V or $V(G)$ and a set of edges denoted by E or $E(G)$.
Every edge is associated with a pair of vertices. (Unordered pair)

Eg $G(V, E)$ where $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$
 $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$



Graph containing parallel edges and self loop.

* Terms Related To Graphs

- (1) Order → Cardinality of the set $V(G)$
- (2) Size → Number of edges in the graph $E(G)$
- (3) Self-loop → An edge starting and ending in same vertex (eg. e_1 above)
- (4) Parallel Edges → Edges associated with the same pair of vertices. (eg. e_5 and e_6)
- (5) Adjacent Vertices → Two vertices are said to be adjacent if there is an edge between them. (eg. v_1, v_2)

6) Adjacent Edges $\rightarrow$ Two non-parallel edges are called adjacent, if they are incident on a common vertex. eg. (e_3 and e_4)

7) Incident $\rightarrow$ An edge is called incident to a vertex if the vertex is the end vertex of the edge.
 e_3 is incident on v_2 and v_3

8) Degree of a vertex $\rightarrow$ It is the number of edges incident on a vertex.

Denoted as $\deg(v)$

eg. $\deg(v_1) = 4$

*) Degree is always a positive number

*) Self loop is counted twice.

9) Minimum & Maximum Degree $\rightarrow$

$$\delta(G) = \min(\deg(v) : v \in V(G))$$

$$\Delta(G) = \max(\deg(v) : v \in V(G))$$

10) Isolated Vertex $\nexists$ no edge is incident on the vertex. eg. v_4
Degree of isolated vertex is 0

11) Pendant Vertex $\nexists$ the degree of the vertex is 1
eg. v_7

* Handshaking Theorem

Let $G(V, E)$ be a graph having n vertices and e edges. Then $\sum_{i=1}^n \deg(v_i) = 2e$

Corollary In a graph G , the number of vertices having an odd degree is always even.

Q A graph has 12 edges, 2 vertices of degree 3, two vertices of degree 4, and other vertices of degree 5. Find number of vertices in the graph.

Let number of vertices = n

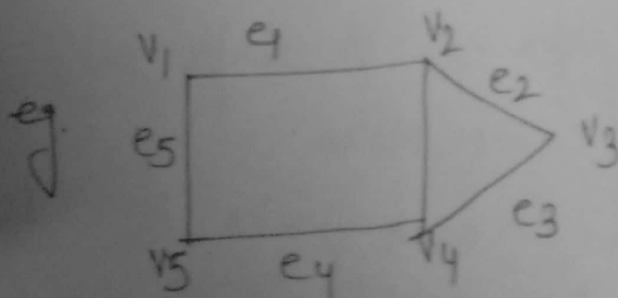
$$2 \times 3 + 2 \times 4 + (n-4) \times 5 = 24$$

$$n-4 = 2$$

$$\underline{n=6}$$

* Types of Graphs $\rightarrow$

1) Simple Graph $\rightarrow$ A graph without self loops and parallel edges.



2) Finite Graph Graph having finite number of edges and finite number of vertices.

3) Multigraph Graph having parallel edges

4) Trivial One vertex and no edge

5) Null Having finite vertices & no edges

• V_1 • V_2
 • V_3
 • V_4

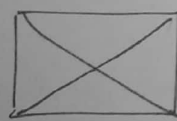
Null Graph

6) Complete Graph $\rightarrow$ If there exists an edge between every pair of vertices.

A complete graph having n vertices is denoted by K_n .



a)



b)

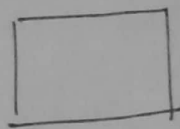


c)

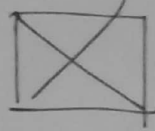
Complete Graphs K_3, K_4, K_5

7) Regular Graph $\rightarrow$ A simple graph with same degree of each vertex.

If degree of each vertex of a regular graph is r , it is called r -regular.



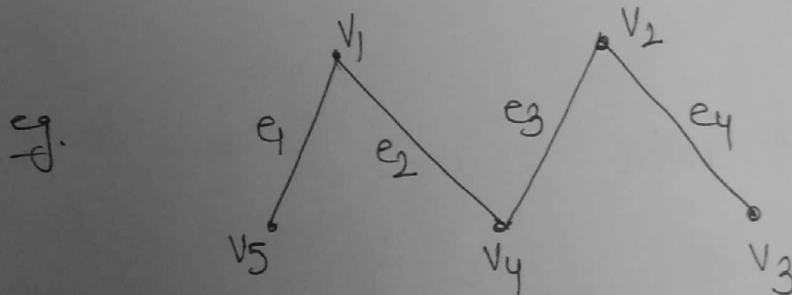
2-regular



3-regular

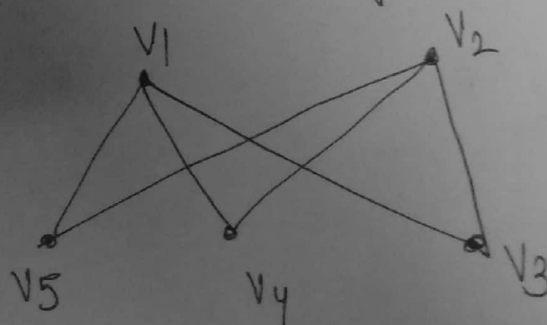
* Every complete graph is regular, but every regular graph is not complete.

8) Bipartite Graph A Graph $G(V, E)$ is said to be a Bipartite Graph, if there exists a partition of the set $V(G)$ into two disjoint sets $V_1(G)$ and $V_2(G)$ such that each edge of the graph has its one end in $V_1(G)$ and other in $V_2(G)$

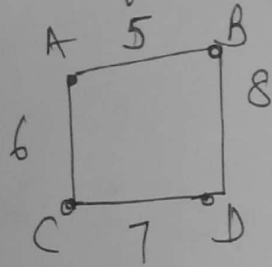


$$V_1(G) = \{v_1, v_2\} \quad V_2(G) = \{v_3, v_4, v_5\}$$

9) Complete Bipartite If each vertex of $V_1(G)$ is joined to each vertex of $V_2(G)$ through an edge.



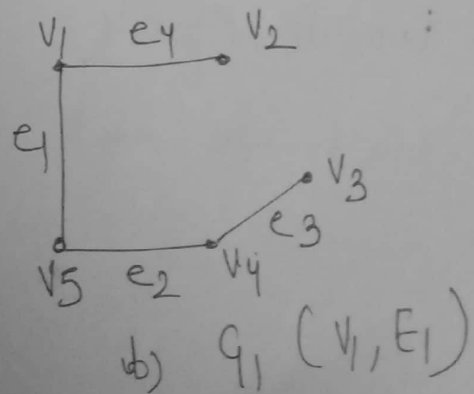
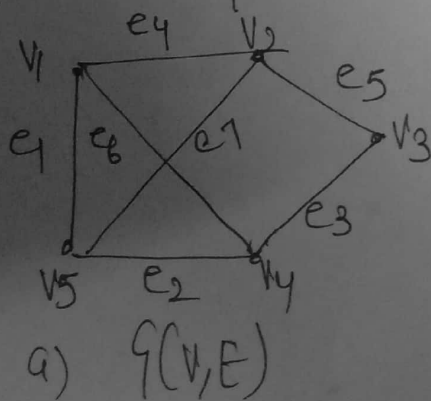
- 10) Weighted Graph $\rightarrow$ If all its edges have been assigned some real number to provide some additional information.



- * There are $\frac{n(n-1)}{2}$ edges in a complete graph.

- * In a Complete Bipartite Graph,
 $e(K_{m,n}) = mn$ where partition is $\{V_m, V_n\}$

- 11) Subgraphs $\rightarrow$ A Graph $G(V_1, E_1)$ is said to be a subgraph of a graph $G(V, E)$ if $V_1 \subseteq V$ and $E_1 \subseteq E$ and each edge has the same end vertices in G_1 as in G .



12)

Directed Graphs Edges have directions

If (u, v) is an edge in directed graph

then $u \rightarrow$ initial vertex

$v \rightarrow$ terminal vertex

in-degree $(v) \rightarrow$ no. of edges with v as their terminal vertex.

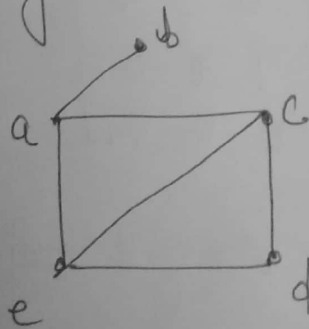
out-degree $(v) \rightarrow$ no. of edges with v as their initial vertex.

* A loop contributes 1 to both in-degree and out-degree of a vertex.

⊗ Representation of Graphs $\rightarrow$

q) Adjacency list $\rightarrow$ which specify the vertices that are adjacent to each vertex of the graph.

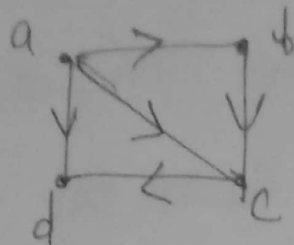
eg. 1



Adjacency list

Vertex	Adjacent Vertex
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

eg 2

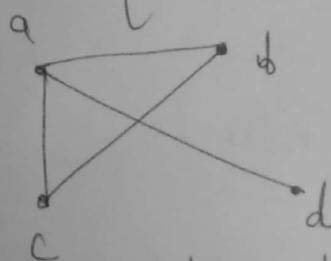


Initial vertex	Terminal vertex
a	b, d, c
b	c
c	d
d	a

b) Adjacency Matrix The Adjacency Matrix $A = [a_{ij}]$
 then

$$a_{ij} = \begin{cases} 1 & \text{if } [v_i v_j] \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

eg:



$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

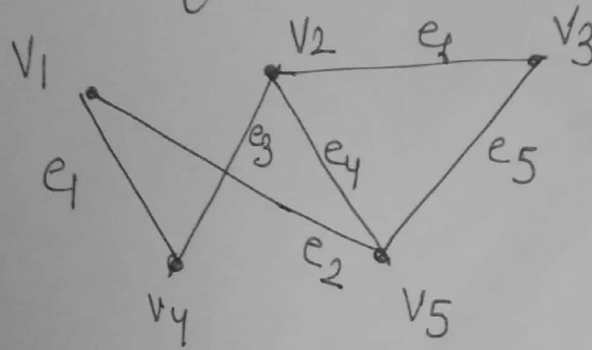
Q Draw a graph with foll. Adjacency Matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

c) Incidence Matrix

The incidence matrix $M = [m_{ij}]$

$$m_{ij} = \begin{cases} 1 & \text{where } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$